THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET C

Chapters 5–8: **Circles, Measuring Space: Perimeter and Area, The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Equal chords of a circle are at _____ distances from the centre. 1

(a) double **(b)** half **(c)** equal **(d)** unequal

2. In a circle of radius 5 cm, a chord is 3 cm from the centre. The length of the chord is: 1

(a) 8 cm **(b)** 10 cm **(c)** 6 cm **(d)** 4 cm

3. A chord of length 14 cm is at a distance of 24 cm from the centre of a circle. The radius of the circle is: 1

(a) 26 cm **(b)** 31 cm **(c)** 14 cm **(d)** 25 cm

4. The length of an arc subtending 90° at the centre of a circle of radius 7 cm ($\pi = \frac{22}{7}$) is: 1

(a) 11 cm **(b)** 5.5 cm **(c)** 22 cm **(d)** 44 cm

5. The area of an equilateral triangle of side a is: 1

(a) $\frac{a^2}{2}$ **(b)** $\sqrt{3}a^2$ **(c)** $\frac{\sqrt{3}}{2}a^2$ **(d)** $\frac{\sqrt{3}}{4}a^2$

6. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 35 cm is: 1

(a) 3850 cm **(b)** 440 cm **(c)** 110 cm **(d)** 220 cm

7. A fair die is rolled once. The probability of getting a number greater than 4 is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{2}{3}$

8. A letter is picked at random from the word ‘PROBABILITY’. The probability of picking the letter B is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{2}{11}$ **(d)** $\frac{9}{11}$

9. The number of outcomes in the sample space when two dice are rolled together is: 1

(a) 36 **(b)** 18 **(c)** 72 **(d)** 38

10. Which term of the sequence $t_n = 3n - 7$ is 332? 1

(a) $n = 115$ **(b)** $n = 114$ **(c)** $n = 113$ **(d)** $n = 112$

11. In the Virahānka–Fibonacci sequence $1, 2, 3, 5, 8, 13, \ldots$ the 7th term is: 1

(a) 34 **(b)** 21 **(c)** 18 **(d)** 20

12. For the sequence $t_n = 4n + 1$, the value of t_{20} is: 1

(a) 80 **(b)** 162 **(c)** 82 **(d)** 81

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. AB is a diameter of a circle and C is a point on the circle. Find $\angle ACB$ and give the reason. 2

14. The diameter of a car tyre is 56 cm. How far does the car travel in 100 complete revolutions of the tyre? ($\pi = \frac{22}{7}$) 2

15. Label as impossible, less likely, even chance, more likely or certain: 2
(i) Monday follows Sunday.
(ii) It snows in Mumbai in July.

16. A sequence is given by $t_1 = -5$, $t_{n+1} = t_n + 3$. Write the first five terms and decide whether 52 is a term. 2

17. Determine whether 97 and 172 are terms of $t_n = 5n - 3$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Prove that the perpendicular from the centre of a circle to a chord bisects the chord. 3

19. Semicircle a on diameter PQ is compared with three semicircles b, c, d on adjacent parts of PQ. Show that the two paths from P to Q are equal in length. 3

20. In a raffle 500 tickets are sold. Riya buys 5 tickets and one ticket wins the prize. What is the probability that Riya wins? 3

21. Find the sum of the first 20 terms of the AP $3, 8, 13, \ldots$ using 'average of first and last × number of terms'. 3

22. Find the area of a circle of radius 21 cm and of the sector of angle 60° ($\pi = \frac{22}{7}$). 3

OR

A track has two straight sections of 84.39 m each and two semicircular ends of radius 36.5 m. Find the length of the innermost edge (use $\pi = 3.1416$). 3

23. If $P(E) = \frac{5}{12}$, find $P(\text{not } E)$ and show that $P(E) + P(\text{not } E) = 1$. 3

OR

Two dice are rolled. Find the probability that 3
(i) the sum is 8
(ii) the sum is at least 10.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. Prove that the area of a parallelogram with base b and height h is bh. What changes if the parallelogram is 'thin'? 5

OR

(i) Verify Brahmagupta's formula for a rectangle with sides a, b. 5
(ii) Verify it for an isosceles trapezium with parallel sides 6 cm and 14 cm and legs 5 cm.

25. (i) Prove that $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. 5
(ii) Hence find S_{20} and S_{50}.
(iii) Find $25 + 26 + \cdots + 58$.

OR

(i) Write the first ten terms of the Virahānka–Fibonacci sequence. 5
(ii) Find the sum of the first 8 terms and compare with V_{10}.
(iii) State where and when it was first written down.

END OF QUESTION PAPER — **Total: 50 Marks**